

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Foundations of Machine Learning | Course Code: 21AI51 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** Calculate the Entropy and Information Gain for selecting the root attribute from a dataset with 9 Yes and 5 No instances, evaluating feature Wind (Weak: 6 Yes, 2 No; Strong: 3 Yes, 3 No) using ID3 Decision Tree algorithm.

[8 Marks]